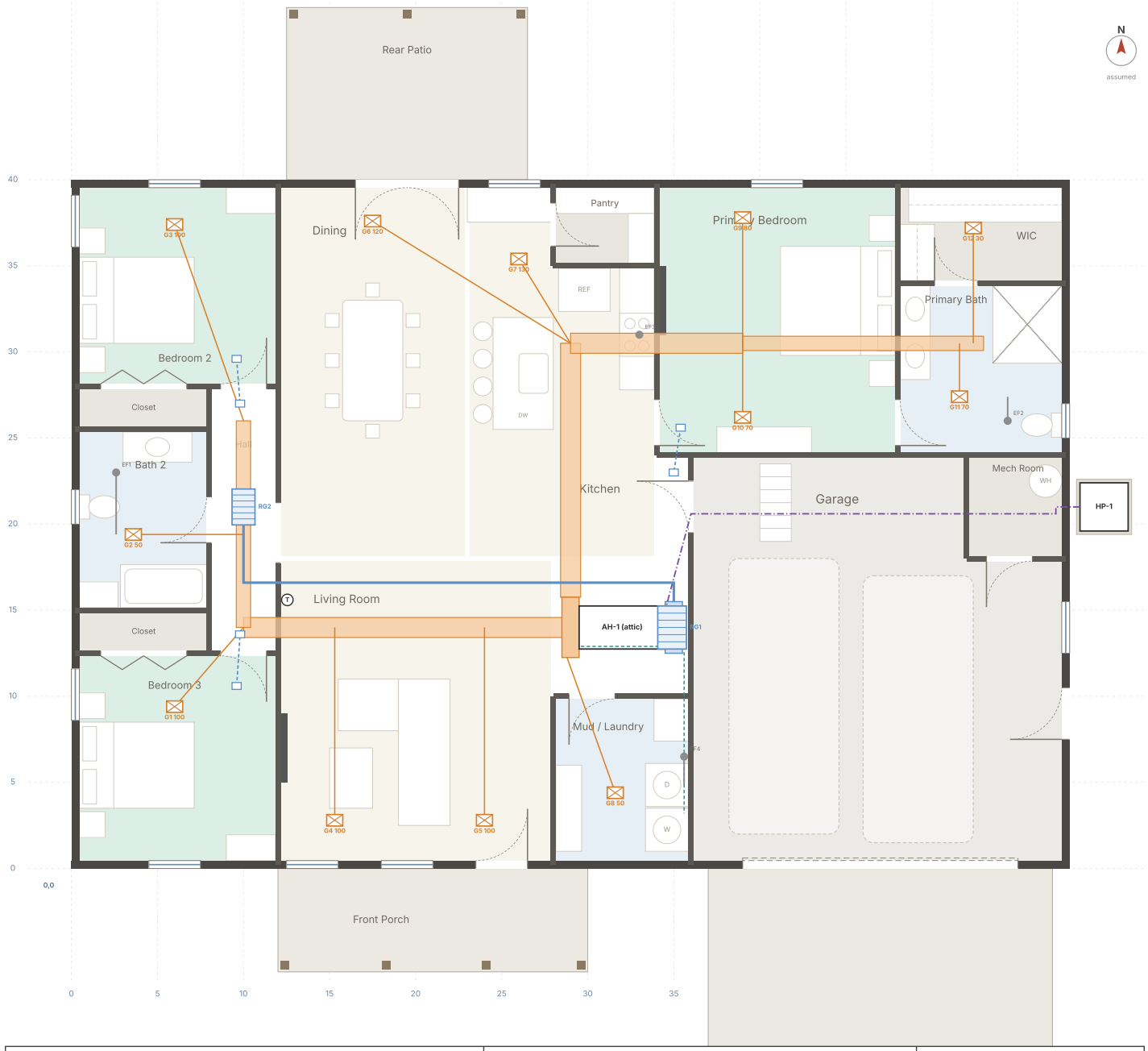


Waring Build BuildWise plan 3933 · 3 bed, 2 bath · 1,800 sq ft + 528 sq ft garage · 58'-0" x 40'-0" framing Slab on grade · 9' ceilings · 6:12 gable · grid 5', 0,0 at the south-west corner	**HVAC plan** Air handler AH-1 in the attic, heat pump HP-1, ducts, registers, exhaust Schematic for pricing and coordination; equipment and duct sizes need a Manual J/S/D.	**M-1** rev drawn for K. Christianson

M-1 HVAC

supply trunks and flex, ceiling diffusers G# returns RG# and jump ducts J# refrigerant line set condensate exhaust ducts EF#

Notes

1. The air handler AH-1 sits in the attic on a platform over the back hall, as the owner asked; the mechanical room is in the garage (water heater, panel, manifold). The way to AH-1 is the pull-down stair in the garage ceiling, with a catwalk and a 30" service space (M1305.1.3), a light switched at the access and a receptacle at the unit.
2. Every duct is in the attic: sheet-metal trunks with R-8 wrap, R-8 flex to ceiling diffusers, every joint sealed with mastic. Keep the flex runs short and straight, and pull them tight. Bury the ducts in the attic insulation where they cross the ceiling.
3. Airflow (1,000 CFM) is shared by room area and exposure for pricing; the HVAC contractor's Manual J (loads), S (equipment) and D (ducts) set the final numbers. Two filtered returns: RG1 under the unit and RG2 in the bedroom hall; the bedrooms return through jump ducts J1–J3.
4. Heat: a cold-climate heat pump with 10 kW of electric strips in AH-1 for backup and defrost. Zone 6 assumed; confirm the design temperature.
5. Ventilation: EF2 in the primary bath runs continuously at 50 CFM to meet IRC M1505.4 whole-house ventilation, boosting to 80 for showers. An ERV is the better choice in a tight cold-climate house (open item).
6. Range hood 300 CFM, ducted 6" to a roof cap; under 400 CFM, so no makeup air is required (M1503.6). Dryer vent rigid 4", 11' developed length with two elbows, roof cap on the front slope.

Open items

The particulars to settle on this sheet.

- Air handler in the attic: in a cold climate, the attic is the coldest place for ducts and the unit. Confirm, or move AH-1 into a conditioned closet (the back hall alcove fits a vertical unit).
- Heat source: heat pump with electric strips assumed. Gas furnace instead?
- Whole-house ventilation: continuous bath fan (drawn) or an ERV (about $2,500–$4,000 installed)?
- Manual J/S/D from the HVAC contractor before ordering equipment; the 2.5-ton size is an estimate for 1,800 sq ft.

Equipment schedule

Tag	Equipment	Where	Size	Notes
AH-1	Air handler	attic over the back hall	2.5 ton, 1,000 CFM, variable speed, 10 kW strips	AH-1: variable-speed air handler with a 2.5-ton cold-climate heat pump coil and 10 kW strips, horizontal on a platform in the attic over the back hall; 30" service platform at the access side, auxiliary drain pan with a float switch
HP-1	Heat pump, outdoor	east side, by the mech room	2.5 ton cold-climate (rated to -13°F or below)	HP-1: 2.5-ton cold-climate heat pump on a raised pad (above the snow line), 18" off the east wall
T1	Thermostat	wall 5, living room	communicating, with outdoor sensor	thermostat on wall 5, living-room side, 54" AFF; the hall return and the cased opening keep it in the house air
EF1	Bath 2 fan		80 CFM	4" duct to a roof cap, rear slope
EF2	Primary bath fan		80 CFM	4" duct to a roof cap, rear slope
EF3	Range hood		300 CFM	6" duct to a roof cap with damper, rear slope
EF4	Dryer vent		—	4" duct to a dryer roof cap, front slope; rigid metal, 11' developed with 2 elbows (M1502.4.5)

Register schedule

1000 CFM supply in all.

Tag	Room	CFM	Flex	Diffuser	Position
G1	Bedroom 3	100	7"	10×6 4-way ceiling diffuser	5'-7" from west ext. wall, 2'-11" from W6
G2	Bath 2	50	6"	8×6 ceiling diffuser	3'-2" from west ext. wall, 4'-3" from W7
G3	Bedroom 2	100	7"	10×6 4-way ceiling diffuser	5'-7" from west ext. wall, 2'-2" from rear ext. wall
G4	Living Room	100	7"	10×6 4-way ceiling diffuser	3'-2" from W5, 2'-4" from front ext. wall
G5	Living Room	100	7"	10×6 4-way ceiling diffuser	3'-10" from W11, 2'-4" from front ext. wall
G6	Dining	120	8"	10×6 4-way ceiling diffuser	5'-4" from W5, 1'-11" from rear ext. wall
G7	Kitchen	130	8"	10×6 4-way ceiling diffuser	1'-10" from W18, 4'-2" from rear ext. wall
G8	Mud / Laundry	50	6"	8×6 ceiling diffuser	3'-5" from W11, 3'-11" from front ext. wall
G9	Primary Bedroom	80	7"	10×6 4-way ceiling diffuser	4'-10" from W15, 1'-9" from rear ext. wall
G10	Primary Bedroom	70	6"	10×6 4-way ceiling diffuser	4'-10" from W15, 2'-1" from W14
G11	Primary Bath	70	6"	10×6 4-way ceiling diffuser	3'-5" from W16, 3'-3" from W14
G12	WIC	30	5"	8×6 ceiling diffuser	4'-3" from W16, 2'-4" from rear ext. wall

Return schedule

Tag	Room	CFM	Size	Notes
RG1	Back Hall	650	20×30 filter grille, MERV 8	central return under AH-1, straight into the return plenum
RG2	Hall	350	16×25 filter grille, MERV 8	bedroom-wing return, 14" insulated flex to the return plenum
J1	Bedroom 3	—	10" jump duct	ceiling grille to the hall, so the room returns with its door shut
J2	Bedroom 2	—	10" jump duct	ceiling grille to the hall, so the room returns with its door shut
J3	Primary Bedroom	—	10" jump duct	ceiling grille to the hall, so the room returns with its door shut

Duct schedule

Trunk	Size	Length	Notes
ST1	14×8	18'-6"	west trunk in the attic, sheet metal, R-8 wrap; reduces to 10×8 past the living room take-offs
ST1b	10×8	12'-0"	bedroom-wing trunk over the hall
ST2	14×8	24'-9"	north trunk over the kitchen
ST2b	10×8	14'-0"	primary-suite trunk

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The HVAC contractor takes off the final list.

Item	Qty	Notes
5" insulated flex, R-8	1 × 25' bags	10' measured +20%
6" insulated flex, R-8	2 × 25' bags	37' measured +20%
7" insulated flex, R-8	4 × 25' bags	69' measured +20%
8" insulated flex, R-8	2 × 25' bags	33' measured +20%
Sheet metal trunk 14×8 / 10×8, R-8 wrap	77'	plus take-off collars with dampers, one per register
Ceiling diffusers	12	with boots, sealed to the drywall

Item	Qty	Notes
Filter return grilles 20×30 and 16×25	2	MERV 8 filters
Jump ducts, 10" flex with 2 grilles each	3	
Bath fans 80 CFM with roof caps	2	EF2 with a continuous-ventilation control
Mastic, UL 181 tape, duct straps	lot	every joint sealed; total leakage ≤ 4 CFM25 per 100 sq ft (IECC R403.3.5)